

Novel Project Proposal: Hybrid Controllable Style-Content Disentanglement Network (HC-SCDNet)

Project Title

"HC-SCDNet: A Hybrid β -VAE and Lightweight Diffusion Architecture for Controllable Style-Content Disentanglement in Neural Style Transfer"

Problem Statement and Motivation

Current neural style transfer approaches face several limitations:

- **Style-Content Entanglement:** Existing methods struggle to independently control style and content aspects^{[37][38]}
- **Limited Controllability:** Most approaches offer binary style transfer without fine-grained control^{[39][40]}
- **Computational Inefficiency:** High-quality results require computationally expensive models^{[41][42]}
- **Reproducibility Issues:** Lack of systematic evaluation frameworks for style transfer quality^[33]

Recent research shows that β -VAEs excel at disentanglement while lightweight diffusion models achieve efficient high-quality generation. However, no work has systematically combined these approaches for controllable style transfer.^{[12][17][24]}

Technical Architecture Overview

Core Innovation: Hybrid architecture combining β -VAE for disentanglement with a lightweight diffusion model for refinement.

Stage 1: Disentangled Encoding (β -VAE Module)

- **Style Encoder:** Extract style-specific latent representations (texture, color, brushstrokes)
- **Content Encoder:** Extract semantic content representations (shapes, objects, composition)
- **Disentanglement Loss:** β -weighted KL divergence to ensure independence between style and content latents

Stage 2: Controllable Fusion (Lightweight Diffusion Module)

- **Conditional U-Net:** Minimal parameter diffusion model (~50M parameters, inspired by ControlNet-XS)^[24]

- **Multi-Scale Attention:** Integrate disentangled representations at different resolution levels
- **Progressive Refinement:** Few-step denoising process (5-10 steps) for efficiency

Stage 3: Quality Enhancement (Adversarial Refinement)

- **Lightweight Discriminator:** Focus on perceptual quality and style fidelity
- **Perceptual Loss:** VGG-based content preservation with Gram matrix style matching

Key Technical Innovations

1. **Disentangled Conditioning:** Use β -VAE latents as conditions for diffusion model, enabling independent style/content control
2. **Efficient Architecture:** Lightweight design achieving <100M total parameters
3. **Progressive Control:** Multi-level conditioning allowing coarse-to-fine style application
4. **Reproducible Framework:** Standardized evaluation protocol with regression testing

Datasets and Evaluation Metrics

Training Data:

- **WikiArt Dataset:** 80,000 artistic images across 27 styles
- **COCO Scenes:** 50,000 content images for diverse semantic content
- **Synthetic Pairs:** Generated style-content combinations for controlled evaluation

Evaluation Metrics:

- **Disentanglement Scores:** β -VAE metric and mutual information gap
- **Style Fidelity:** Gram matrix distance and learned perceptual image patch similarity (LPIPS)
- **Content Preservation:** SSIM and semantic segmentation consistency
- **Efficiency Metrics:** Inference time, memory usage, and parameter count
- **Controllability Assessment:** Human evaluation on independent style/content manipulation

Implementation Tools and Libraries

Core Framework:

- **PyTorch Lightning:** For reproducible training pipeline

- **Diffusers Library:** Base diffusion model implementations
- **Weights & Biases:** Experiment tracking and reproducibility
- **TensorBoard:** Real-time monitoring and visualization

Specialized Libraries:

- **β -VAE Implementation:** Custom PyTorch modules based on recent disentanglement research^[12]
- **Style Transfer Utils:** Pre-processing and evaluation utilities
- **Reproducibility Tools:** Version control for datasets and model checkpoints^[33]

Expected Deliverables (2-Week Timeline)

Week 1: Foundation and Basic Implementation

- **Day 1-2:** Literature review completion and architecture design finalization
- **Day 3-4:** Dataset preparation and preprocessing pipeline
- **Day 5-7:** β -VAE module implementation with disentanglement losses

Week 2: Integration and Evaluation

- **Day 8-10:** Lightweight diffusion module implementation and integration
- **Day 11-12:** Training pipeline setup and initial model training
- **Day 13-14:** Evaluation framework implementation and preliminary results

Concrete Outputs:

1. **Working Prototype:** Functional HC-SCDNet model with basic style-content disentanglement
2. **Evaluation Framework:** Comprehensive metrics suite with reproducible evaluation protocol
3. **Benchmark Results:** Quantitative comparison with 3-5 baseline methods (NST, AdaIN, StyleGAN-based approaches)
4. **Code Repository:** Well-documented, reproducible codebase with training scripts
5. **Technical Report:** Detailed methodology, preliminary results, and future directions (8-10 pages)

Expected Outcomes and Impact

Immediate Contributions:

- First systematic combination of β -VAE disentanglement with efficient diffusion models for style transfer
- Lightweight architecture suitable for real-time applications
- Reproducible evaluation framework for style transfer research

Broader Impact:

- **Academic:** Novel hybrid architecture approach applicable to other generative modeling tasks
- **Industry:** Efficient model suitable for mobile and edge deployment
- **Creative Tools:** Enhanced controllability for digital art and content creation applications

Feasibility Analysis

Technical Feasibility: High - builds on well-established components (β -VAE, diffusion models) with clear integration path

Resource Requirements: Moderate - can be implemented with single GPU (RTX 3090/4090) within timeframe

Risk Mitigation: Incremental development approach allows for quick pivoting if integration challenges arise

Evaluation Feasibility: Strong - comprehensive evaluation possible within timeframe using established metrics

This project addresses current research gaps while providing practical value, combining academic rigor with implementable innovation suitable for a 2-week development cycle.